ESAS LOOKSFAM
Compiled by: Kyo

* Ratio of shear stress to the unit deformation in shear = Shear Modulus
* Unit shear stress divided by unit shear deformation = modulus of rigidity
* Law expressed relationship between stress- strain = Hooke's Law
* Inelastic (plastic) failure of the beam = Rotation
* Load is not applied through the centroid = Eccentric Load
* Centroid of tension or compression member's section = Axial Load
* Perpendicular to the force = Normal Load
* Loading does fatigue occur = Repeated Load
* Length per degree rise in temperature = Linear expansivity
* Stress at failure = Rupture Strength
* Unit lateral deformation to the unit longitudinal deformation = Poisson's Ratio
* Stress to strain below proportional limit = Young's ratio
* Stress strain diagram – slope of the curve = modulus of elasticity
* Obtained by repeatedly loading a specimen at stress levels until it fails = Endurance Limit or fatigue strength or material
* Stress which material is capable withstanding without deviation = Proportional Limit
* Endure an infinite number of stresses cycles = Endurance limit
* Deviation from edge straightness = camber
* Nominal stress at fracture in a tension at constant load and constant temperature = Stress rupture strength
* Financial ratio indicates company's current assets will be sufficient to meet company's obligation = Liquidity ratio
* RA 7832 = Anti electricity Pilferage Act
or Pilferage of electricity theft , theft of electric power transmission line materials rationalizing system losses as a component thereof and for the purpose.
* Old EE law = RA 184
* PRC modernization = RA 8981
* Electronic device can read output vertical lines into language = bar code reader
* Allowing foreign engineer to take board exam = Foreign reciprocity
* Peace and order problem exist = Security red alert / security alert
* Property or material changes when amount of substance changes = Extensive
* DO NOT Change = Intensive
* EPIRA is divided into four sectors ;
o 1. Generation
o 2. Transmission
o 3. Distribution
o 4. End-user
* The motor disconnection means must be near to the motor protected = Installation rule
* Murang Kuryente Act = RA 11371
* Field practice of REE is = Sec 31 (b)
* CPD – Sec 21
* Slender member which prevents parts of the structure moving together = STRUT
* Reynolds number = R=I/V or RIV, Inertial force/viscous force

* Female port 3 and 15 pins = VGA port
* Involved in bronze the alloy of copper and tin = Metallic bond
* Ionic and covalent = H2
* FM transmitter is proportional to = Amplitude
* 0.6% carbon that has been spheroidized cold drawn = Cutting Tool
* Cold working = Increased Strength Decreased Ductility
* Grounded neutral terminal on a polarized receptable shall be by metal coating = White in color
* Assets to cash = Liquidity
* Effective and safe means of releasing water by use of = Spillway
* Nuclear energy = Fission
* ¬IRR of RA 7920 approved under Board Resolution no. = 18
* Code of Ethics for electrical engineering was adopted by the Board resolution no. 22 in = October 18,1995
* It is a duty of electrical engineer to involve himself in activities designed to promote welfare = Code of ethics
* RA 7920 was approved by Fidel V Ramos – February 24,1995
* Basic Accounting Equation = Assets = Liabilities + Owners Equity
* Maximum shear stress for triangular beam = 3s/bh
* Commercially used pressure measuring device = Bourdon Pressure Gage
* Cast Iron which CANNOT machined = White Cast
* Two-way mode of communication =FULL DUPLEX
* Either party can hear while talking = Full Duplex
* Only one to two party can transmit at a time = Half duplex
* OSHS for electrical safety = 1210

Yellow Alert = Low contingency reserve than required
Red Alert = Contingency reserve is zero, there is generation deficiency, critical loading or imminent overloading
Blue Alert = Tropical Disturbance about to land fall in about 24 hours
Security Alert = There is peace and order problem
Demand Control Imminent Warning = Load Shedding in the next 30 minutes

*Regnault's Law = Specific heat
*Bernoulli's Equation = Conservation of Energy
*Kelvin –Planck's Law = Conversation of Work into Heat
*Varignon's Theorem = Location of Centroid
*D'Alembert's Theorem = Newton's First law of Motion

Newton's Law of Motion
*Newton's First law = Law of Inertia
*Newton's Second law = Law of Acceleration
*Newton's Third law = Reaction

Laws of Thermodynamics
*First Law = Energy
*Second Law = Enthrophy
*Third Law = Absolute Zero Temperature
*Zeroth Law = Equilibrium

VIOLET = Shortest
RED = Longest
INDIGO = Smallest

Humans Body
Oxygen = 65%
Carbon = 18%
Hydrogen = 10%
Other = 7%

REYNOLDS NUMBER
*Laminar Flow = Less than 2300
*Critical Flow = 2300 – 4000
*Turbulent Flow = more than 4000

ASTM
*D732 = Shear Strength
*D790 = Bending Strength
*D638 = Tensile Strength
*D695 = Compressive Strength

Colinear = Lie on the same point
Coplanar = Lie on the same plane
Concurrent = Meet at the same point

POLYTROPIC EXPONENT, n
*n=0 ; ISOBARIC
*n=1 ; ISOTHERMAL
*n=k ; ISENTROPIC
*n=∞ ; ISOMETRIC

CH20 – Empirical Formula of Glucose
CH3 – Empircial Formula of C2H6
NH3 – Chemical Formula of Ammonia
CACL2 – Calcium Chloride
A12(S04)3 – Aluminum Sulfate

LOADS
Concentrated – WL / 4
Uniform Load – WL^2 / 8
Total Load – WL / 8

Economics
***Monopoly = One Seller , Many Buyer
*Monopsony = Only One Buyer
*Oligopoly = Few Seller, Many Buyer
*Oligopsony = Many Seller , Few Buyer
*Bilateral Monopoly =
- One Buyer, One Seller
*Bilateral Oligopoly =
- Few Seller, Few Buyer
*Doupsony = Many Seller, Two Buyer
*Perfect Competition =
- Many Buyer, Many Seller

Direct = 90% , Downward
Indirect = 90% , Upward
Semi-Direct = 60%-90%, Downward
Semi-Indirect = 60%-90%, Upward

IRR – RA 7920

Rule 2: Definition of Terms

(a). **Syllabi**
(b). **Commission** = Professional Regulation Commision
(c). **Commissioner** = Chairman of PRC
(d). **Board** = Board of EE
(e). **CPE** = Continuing Professional Education
(f). **RA 184** = Old EE Law
(g). **PEE** = Professional Electrical Engineer
(h.) **ASSO EE** = Associate Electrical Engineer
(i). **ASST EE** = Assistant Electrical Engineer
(j). **REE** = Registered Electrical Engineer
(k). **ME** = Master Electrician
(l). **RME** = Registered Master Electrician
(m). **IIEE** = Institute of Electrical Engineer
(n). **DECS** = Department of Education Culture and Sports
(o). **CHED** = Commission of Higher Education
(p). **TESDA** = Technical Education and Skills Development
(q). **S.O. No. (B)** = Special Order Number

Safety Man = OSHA Rule 1047
10 – 20K = 7832 States That the Penalty of a Prison Mayor
50-100K = Penalty Redusion Temporal
40 years = PRC Commissioner or chairperson age.

According to RA 8981 chairperson, the commissioner pre must have atleast **5 years** of management experience.

According to section 7 the first person member of CPD council shall be **President/Officer** accreated integrated professional organization duly to authorized by the board of governor.

EE LAWS

RA 184 = Old EE Law
RA 7920 = New EE Law
RA 8981 = PRC Modernization
RA 7832 = Anti – Electricity & Transmission Line / Material Pilferage Act
RA 10912 = Continuing Professional Development Act of 1992
RA 6395 = National Power Corporation
RA 7368 = Country Wide Industrilization Act of 1992
RA 9513 = National Transmission Corporation or Renewable Energy Law
RA 9155 = DECS
RA 7796 = TESDA
RA 7722 = CHED
EO No. 172 = Energy Regulatory Board
EO No. 127 = Department of Finance
PD No. 269 = Electric Cooperative / NEA Created Through
Proclamation No. 193 = Month of **May** as Electrical Safety Month
PD 223 = The Professional Regulations Commission (PRC) Was Created
1973 = Year Created the Professional Regulation Commission (PRC)
RA 11371 = Newly Signed Law of President Duterte called **Murang Kuryente Act**
DOLE = PRC under before
11063 = House Bill
1766 = Senate Bill
RA 9136 = Electric Power Industry
RA 10912 = Signed by into law by Benigno Aquino III (July 21, 2016)

- Adopted an as amendment to the House bill No. **6423**
- And Senate Bill **2581**

Rafael F. Florentino = Old Chairman
Francis V. Mapile = New Chairman
Jaime V. Mendoza = Board Member

First period – 2 elements
Second & Third period – 8 elements
Fourth & Fifth period – 18 elements
Sixth & Seventh – 32 elements
Group 1 – Alkali metals
Group 2 – Alkaline earth metals
Group 15 – Pnictogens
Group 16 – Chalcogens
Group 17 – Halogens
Group 18 – Noble gases

Lithosphere (Earth's surface)
Troposphere 0 – 12km (weather)
Stratosphere 12 – 50km (exists primarily)
Mesosphere 50 – 80km
Thermosphere 80 – 700km
Exosphere 700 – 10,000km

1 Byte = 8 Bits
1KB = 1024 Bytes
1MB = 1024 KB
1GB = 1024 MB
1TB = 1024 GB
1PT = 1024 TB

TW and UF – 60 deg C
THWN – 75 deg C
V – 85 deg C
THHN – 90 deg C

Math – 25%
Esas – 30%
EE – 45 %

18 years old – RME
21 years old – REE
35 yers old – PEE

Computer Symbols and Functions

OVAL – Start / Stop
Parallelogram – Input / Output
Rectangle – Processing
Diamond – Decision Making
Arrows – Flow of Control
Circle – Connector

COLOR CODING (OSHA)

RED – Fire Prevention And Protection Equipment
Black & White – Housekeeping and Aieste Marking
Magenta and Yellow – Radiation and Hazards (X-Ray, Gamma, Beta)
Green – Safety and First Aid
Blue – Defective Machinery / Under Repair
Orange – Warning, Alert (Parts of Machinery or Energized Equipment that may cut, crush)
Yellow – to designate caution and for making physical hazards

Church – 8VA /m^2
Hospital – 16 VA/m^2
School – 24 VA/m^2
Building – 28VA/m^2

ACRONYMS

AAC – All Aluminum Conductors
AAAC – All Aluminum Alloy Conductors
ACAR – Aluminum Conductors Alloy Reinforced
ACSR – Aluminum Conductors Steel Reinforced
AISI – American Iron and Steel Institute
RMS – Root Mean Square
ROA – Return on Assets
RTU – Remote Terminal Unit
IOT – Internet of Things
VOIP – Voice Over Internet Protocol
PVC – Polyvinyl Chloride
STP – Sewage Treatment Plant
PEC – Philippine Electrical Code
PDC – Philippine Distribution Code
PGC – Philippine Grid Code
NEC – National Electrical Code
NEA – National Electrification Administration
DOE – Department Of Energy
PEP – Philippine Energy Plan
NREP – National Renewable Energy Program
DMC – Distribution Management Community
PDA – Personal Digital Assistant
CREM – Competitive Retail Electricity Market
EPIRA – Electric Power Industry Reform Act
JCPC – Joint Congressional Power Commission
SAE – Society of Automotive Engineers
SGC – Successor Generation Company
SOLR – Supplier of Last Resort
WESM – Wholesale Electricity Spot Market
RES – Retail Electricity Supplier
SPUG – Small Power Utilities Group
TSC – Transition Supply Contract
OARC – Open Access and Retail Competition
SAIFI – System Average Interruption Frequency Index
SAIDI – System Average Interruption Duration Index
CPD – Continuing Professional Development
BEE – Board of Electrical Engineering
CPE – Continuing Professional Education
PEE – Professional Electrical Engineer
REE – Registered Electrical Engineer
RME – Registered Master Electrician
IIEE – institute of Integrated Electrical Engineers
DECS – Department of Education, Culture and Sports
CHED – Commission of Higher Education

TESDA – Technical Education Skills Development Authority
TTL – Transistor-transistor Logic
DTML – Diode Transistor Micro Logic
PEMC – Philippine Electricity Market Corporation
PSALM – Power Sector Assets and Liabilities Management
DCTL – Direct Coupled Transistor Logic
DTL – Diode Transistor Logic
AT – Attention
ALU – Arithmetic Logic Unit
ASCII – American Standard Code for Information Interchange
BCD – Binary Code Decimal
EBCDIC – Extended Binary Coded Decimal Interchange Code
SCADA – Supervisory Control and Data Acquisition
SD RAM – Synchronous Dynamic Random Access Memory
BIOS – Basic Input Output System
IBM – International Business Machine Corporation
WIFI – Wireless Fidelity
PPPD – Preliminary Project Planning Data
CPU – Central Processing Unit
CD ROM – Compact Disk Read Only Memory
CMOS – Complementary Metal Oxide Semiconductor
CSS – Cascading Style Sheets
HTML – Hypertext Mark-Up Language
HTTP – Hypertext Transfer Protocol
JPEG – Joint Photographic Experts Group
PNG – Portable Network Graphics
FTTH – Fiber to the Home
LED – Light Emitting Diode
USB – Universal Serial Bus
UPS – Uninterruptible Power Supply
WWW – World Wide Web